



Predictive Care for Modern EV Drivers

Designing a personalized platform that helps EV drivers monitor vehicle health and prevent unexpected failures

UX Case Study | Web Platform

Role: UI/UX Designer

Duration: 4 Weeks





Problem

As electric vehicles grow in popularity, many new EV adopters are interested in sustainable mobility but lack the experience to properly manage and maintain their vehicles. Users often feel uncertain about battery health, charging habits, and maintenance needs, which can lead to inefficient vehicle care and overlooked warning signs.

Goal

Design a personalized EV care platform that helps users easily understand vehicle health, monitor charging behavior, and receive clear guidance for maintaining their EV.



Solution

Develop EVCare, a platform that provides a vehicle health dashboard, battery and efficiency metrics, charging insights, predictive service alerts, weekly summaries, and personalized EV care recommendations to help users confidently manage their vehicles.

User Persona



KARTHIK

Accounts Executive

Age: 30 years
Education: B.E.ECE Engineering
Hometown: Chennai
Status: Living alone
Occupation: Working in a MNC

“I’m ready for the future of mobility, but I need clarity to drive it with confidence.”

About

- Recently purchased his first electric vehicle
- Interested in sustainable mobility
- Needs simple explanations for EV data

Personality

- Practical and cautious
- Curious about new technology
- Prefers simple tools over complex systems

Goals

- Understand battery health clearly
- Develop better charging habits
- Maintain EV for long-term use

Frustrations

- EV dashboards show unfamiliar metrics
- Charging impact on battery is confusing
- Maintenance guidance is unclear

Needs

- Clear vehicle health dashboard
- Charging behavior insights
- Predictive maintenance alerts

Story Board



Curious About EV's



Initial Excitement



Facing Confusion



Finding EVCare Platform



Confident EV Owner



Design Insights



Simplify EV Health Data

Translate complex battery metrics and technical jargon into easy to understand visuals and indicators

Insights by Users

Users find battery data confusing and hard to interpret

Guide Charging Behaviour

Provide personalized insights and recommendations to help users adopt smarter charging habits

Insights by Users

Users are unsure about ideal charging practices and timings.

Enable Predictive Maintenance

Use data to predict potential issues and notify users before problems impact their vehicle

Insights by Users

Users want early affects and reminders for maintenance

Provide Personalized Insights

Deliver weekly summaries ,comparisons and actionable tips tailored to each user's driving pattern

Insights by Users

Users value personalized guidance relevant to their usage

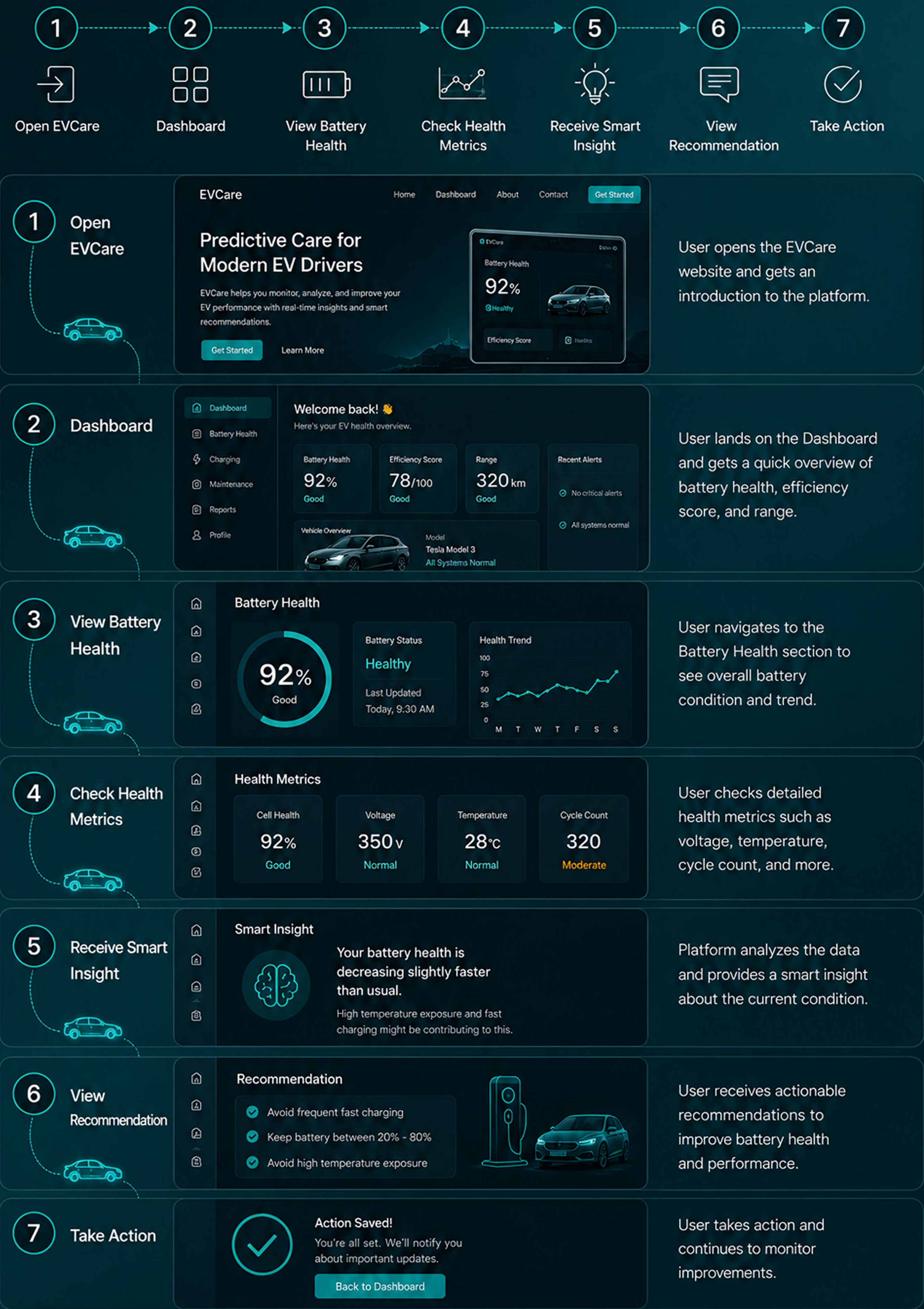
Information Architecture

EVCare is structured around helping users understand their EV's health and take action with confidence.



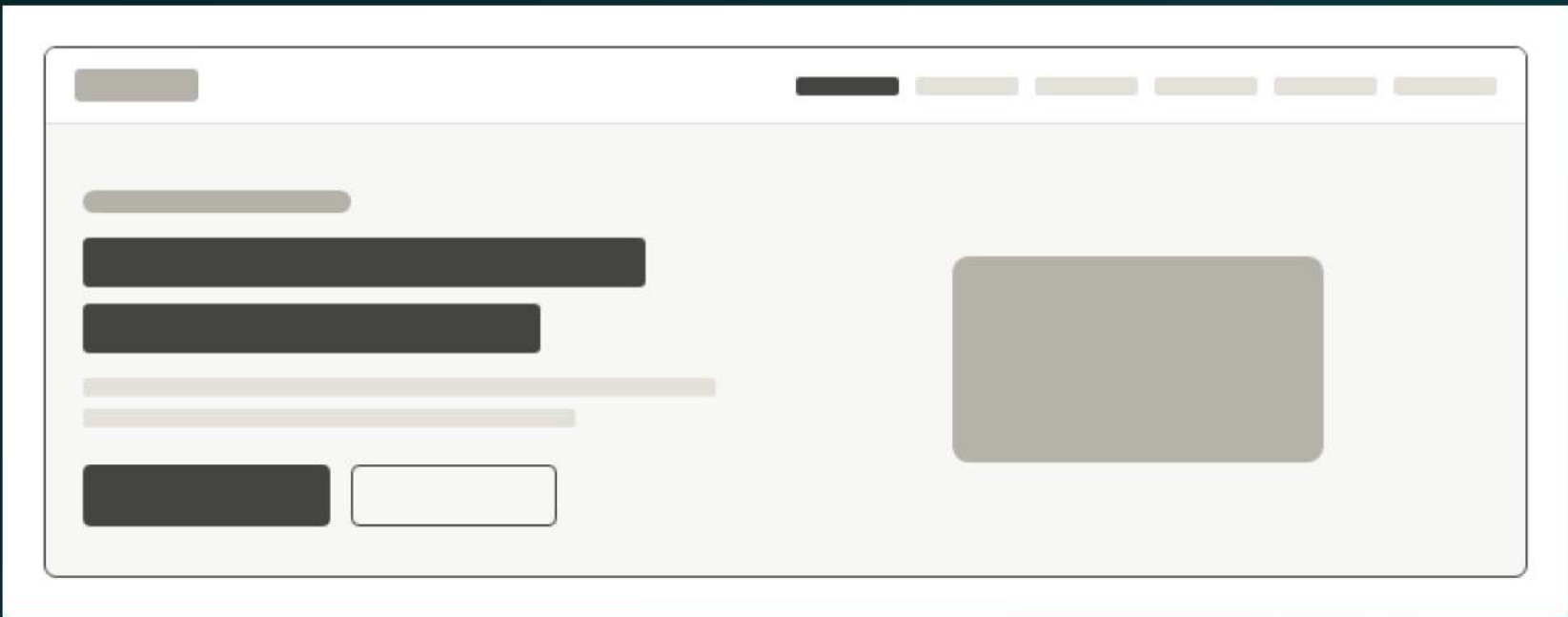
User Flow

This flow illustrates how a new EV user can easily monitor their vehicle health and take action using EVCare.

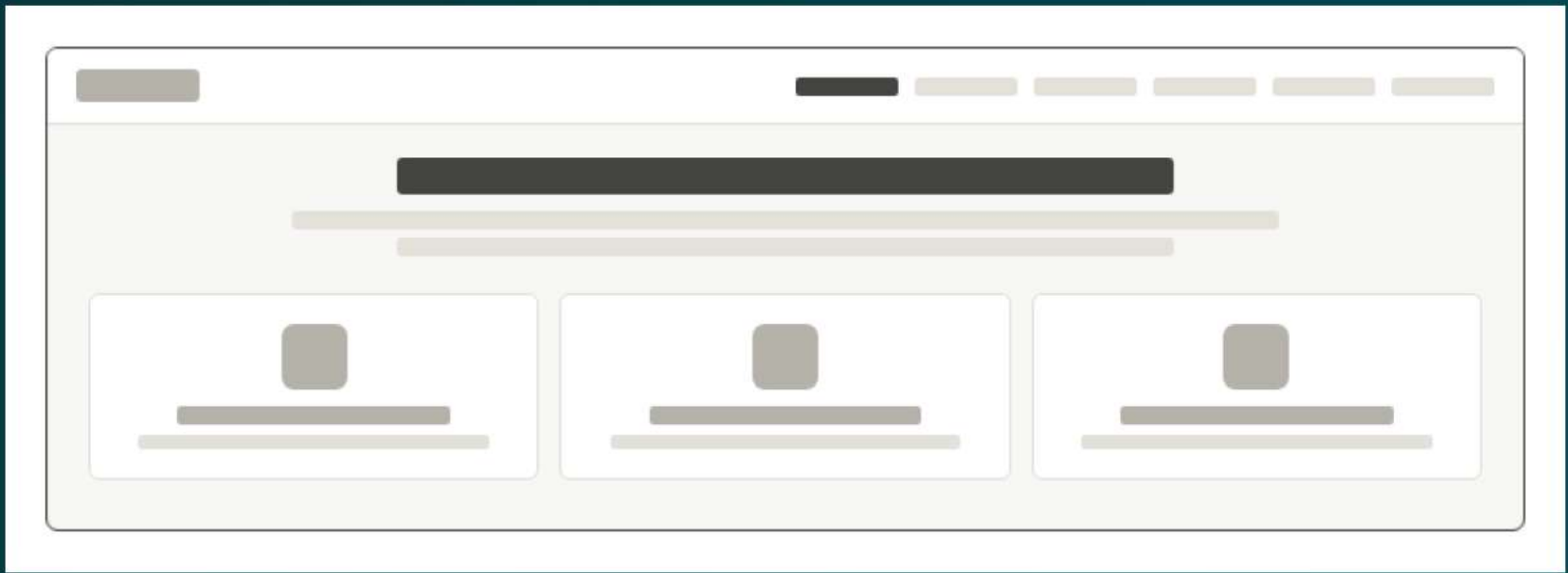


Low Fidelity Wireframes

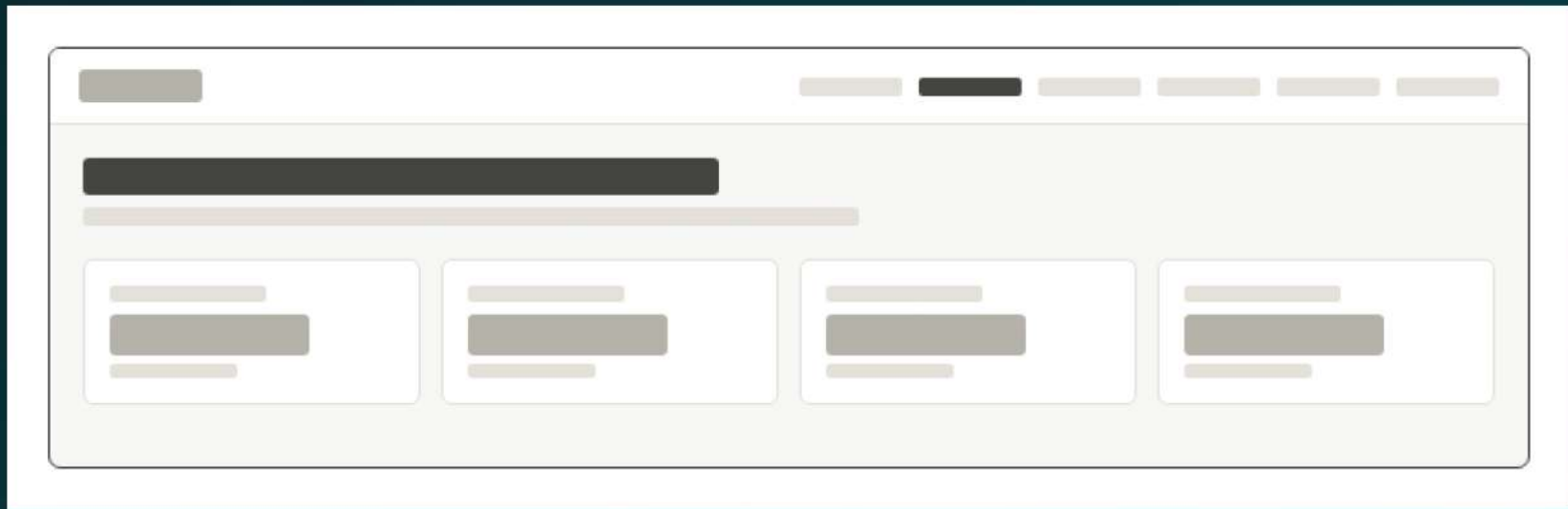
1. Home - Hero Section



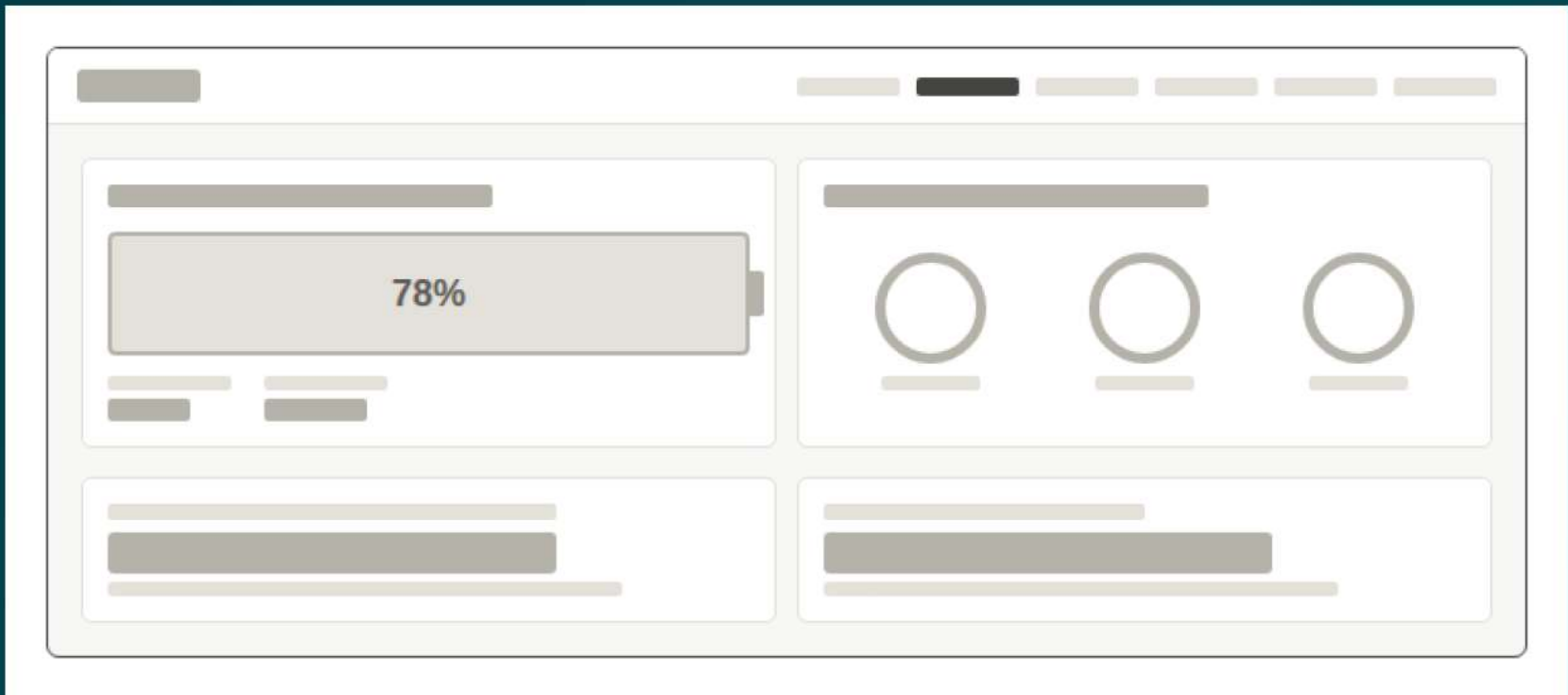
2. Home - Features



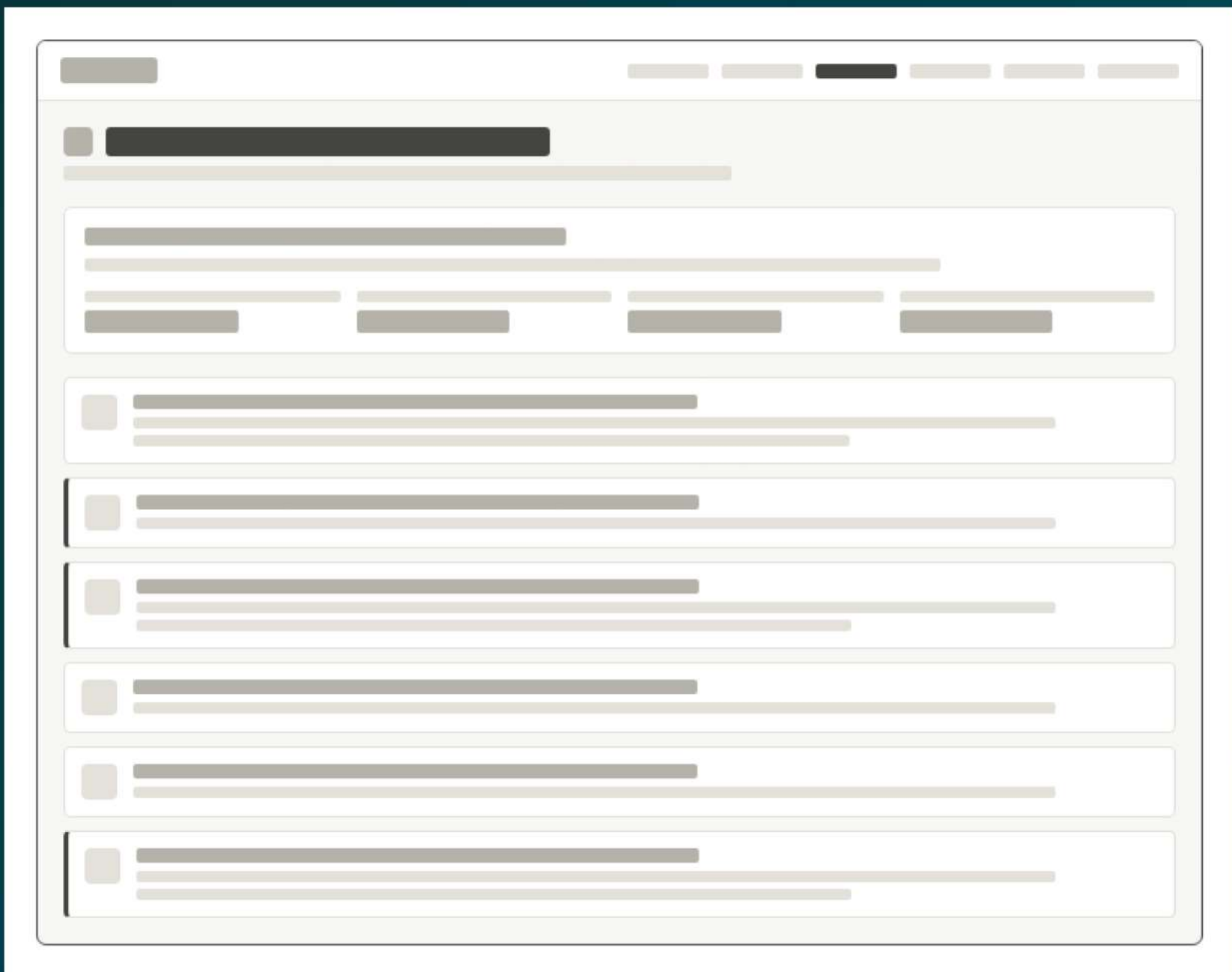
3. Dashboard - Statistics



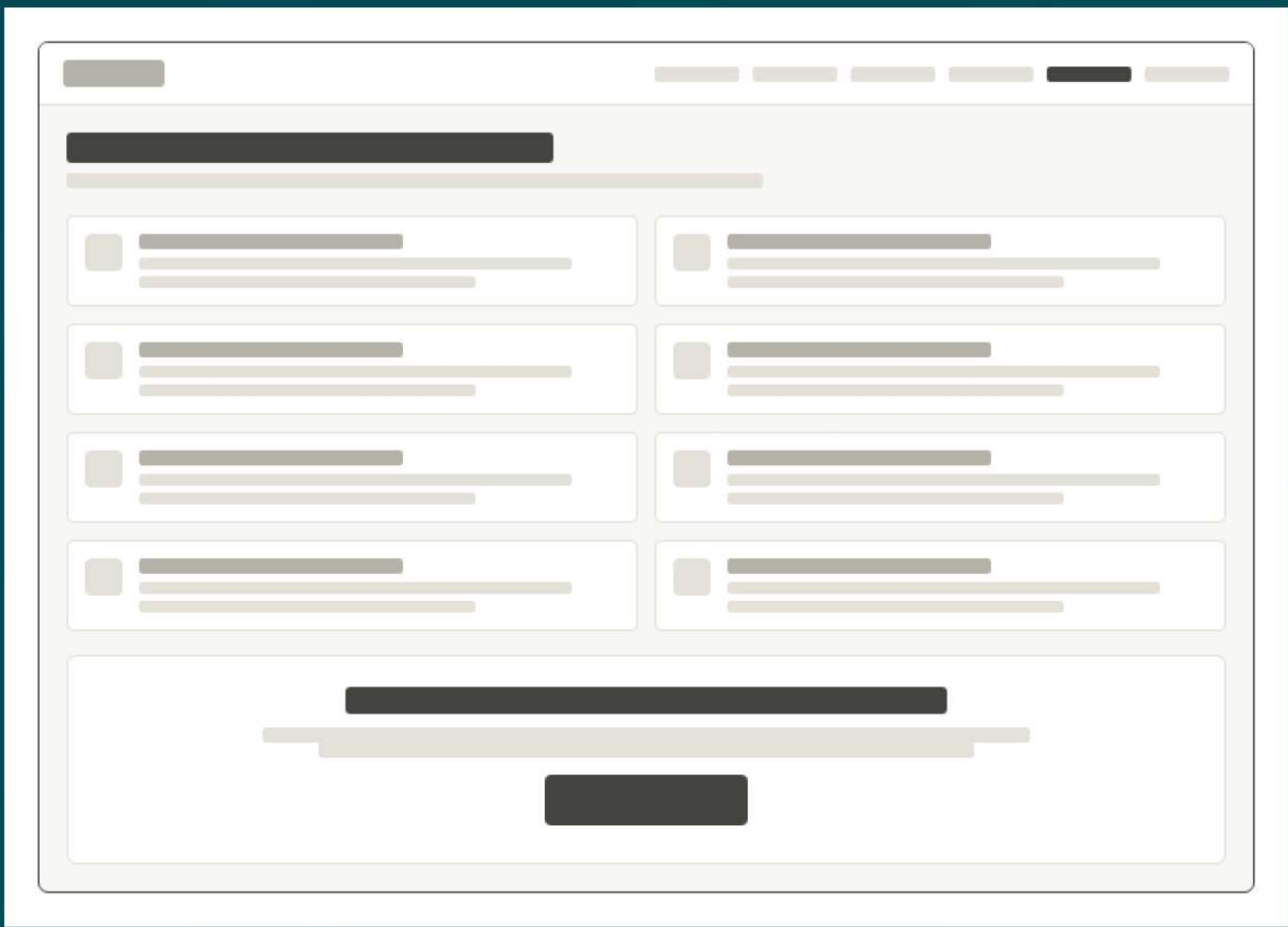
4. Dashboard - Battery Health



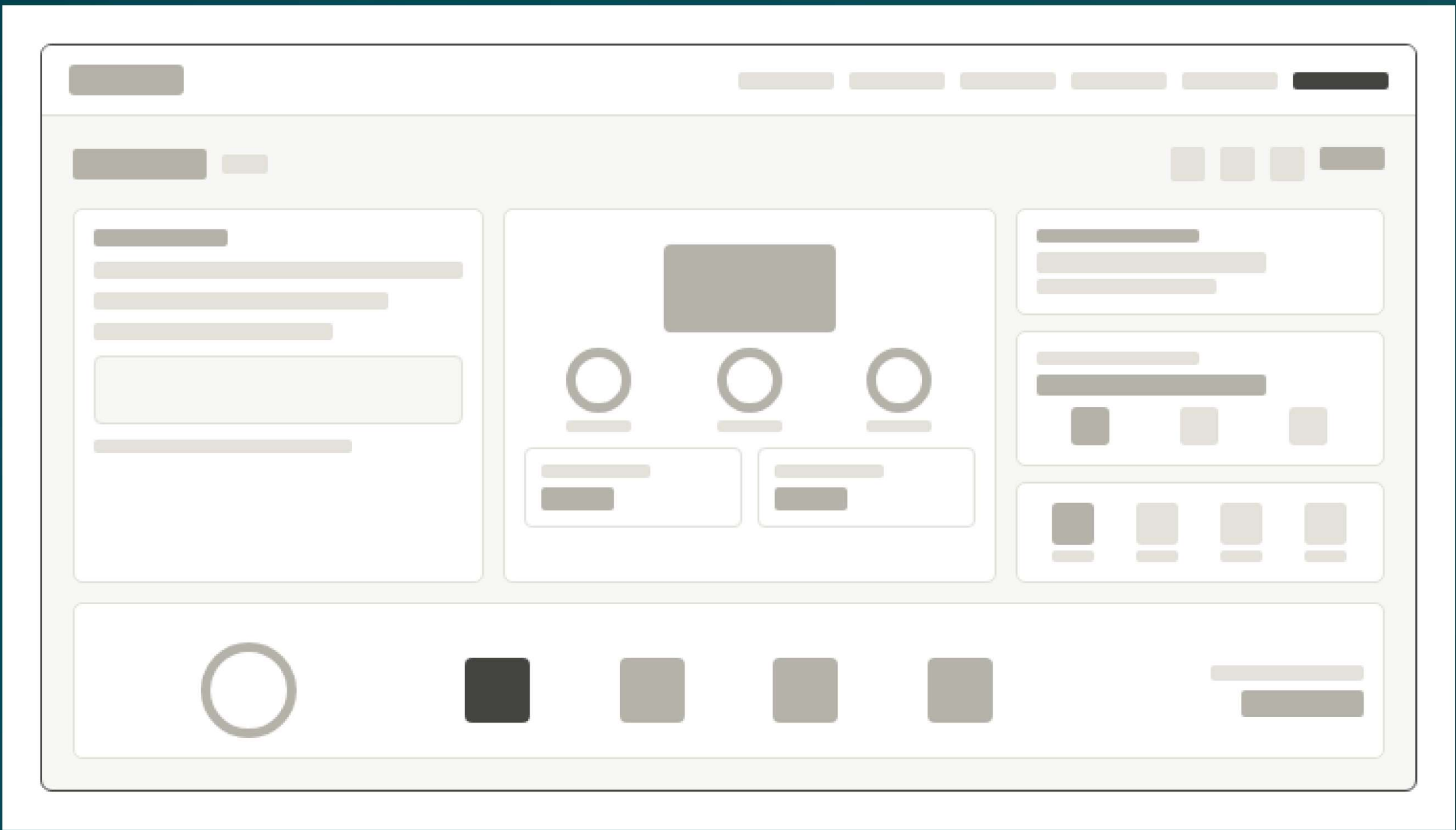
5. Insights - Smart Insights



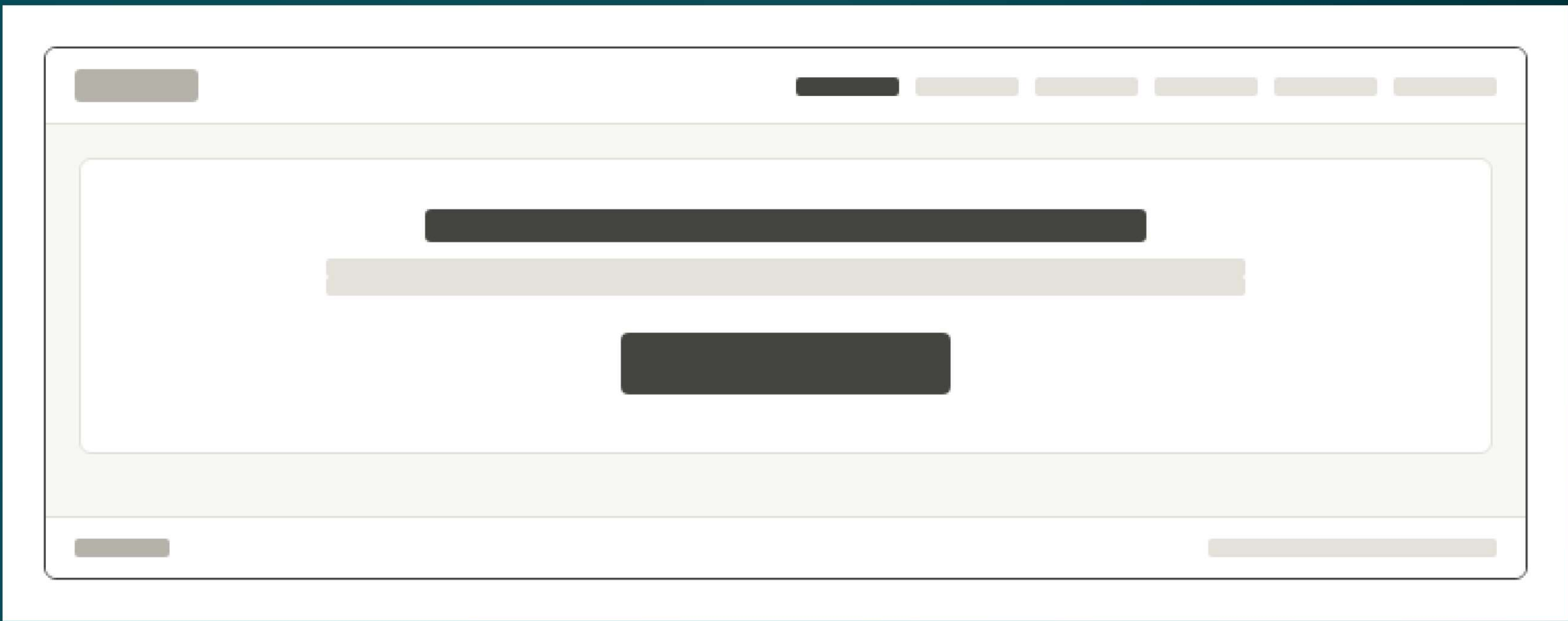
6. Tips - Smart Tips



10. Infotainment



11. CTA - Footer



Design System

A consistent visual language designed to make EV care simple, trustworthy and actionable.

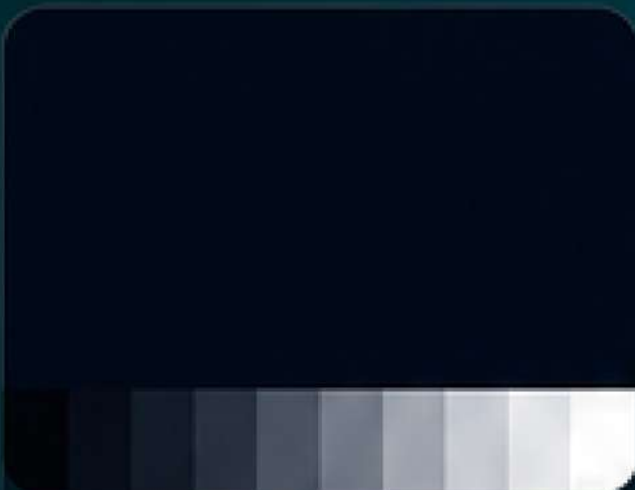


01 COLOR PALETTE



Primary

#00E5FF



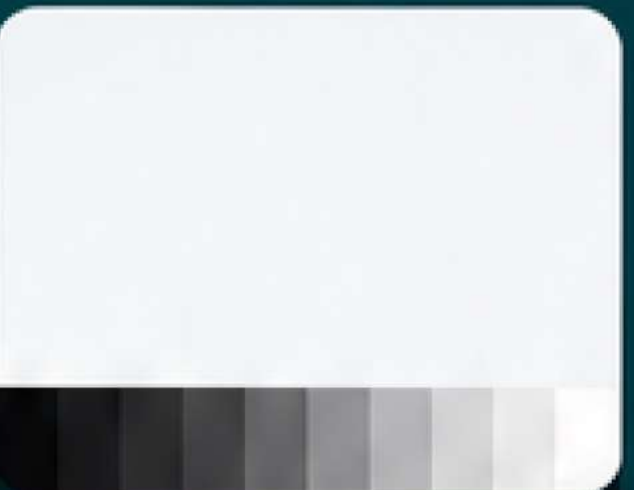
Secondary

#020817



Tertiary

#10B981



Neutral

#F8FAFC



Gray

#6B7280

02 TYPOGRAPHY

H1

Poppins SemiBold
48px / 56px

The quick brown fox jumps over the lazy dog.

H2

Poppins Medium
32px / 40px

The quick brown fox jumps over the lazy dog.

Body

Poppins Regular
16px / 24px

The quick brown fox jumps over the lazy dog. This is body text used for descriptions, supporting text and longer content.

Caption

Poppins Regular
12px / 16px

This is caption text used for labels, hints and less prominent information.

03 COMPONENTS

Primary Buttons

Default



Hover

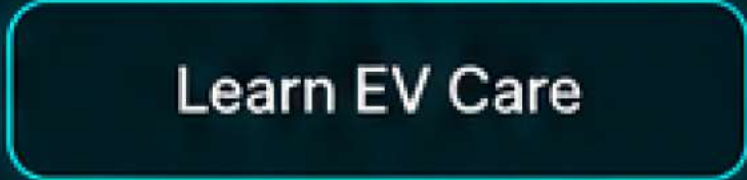


Secondary Buttons

Default



Hover



Dashboard Card

Total Range

320 km



↑ 12% from last week

Battery Health Card

Battery Health



Good

Updated 2h ago

Alert Card



Attention

Schedule service recommended

View Details →

04 ICON SET



Battery



Charging



Diagnostics



Safety



Insights



Profile



Notifications

05 DESIGN PRINCIPLES



Simplicity

Make complex EV data easy to understand and act upon.



Clarity

Present information in a clear, organized and uncluttered way.



Proactivity

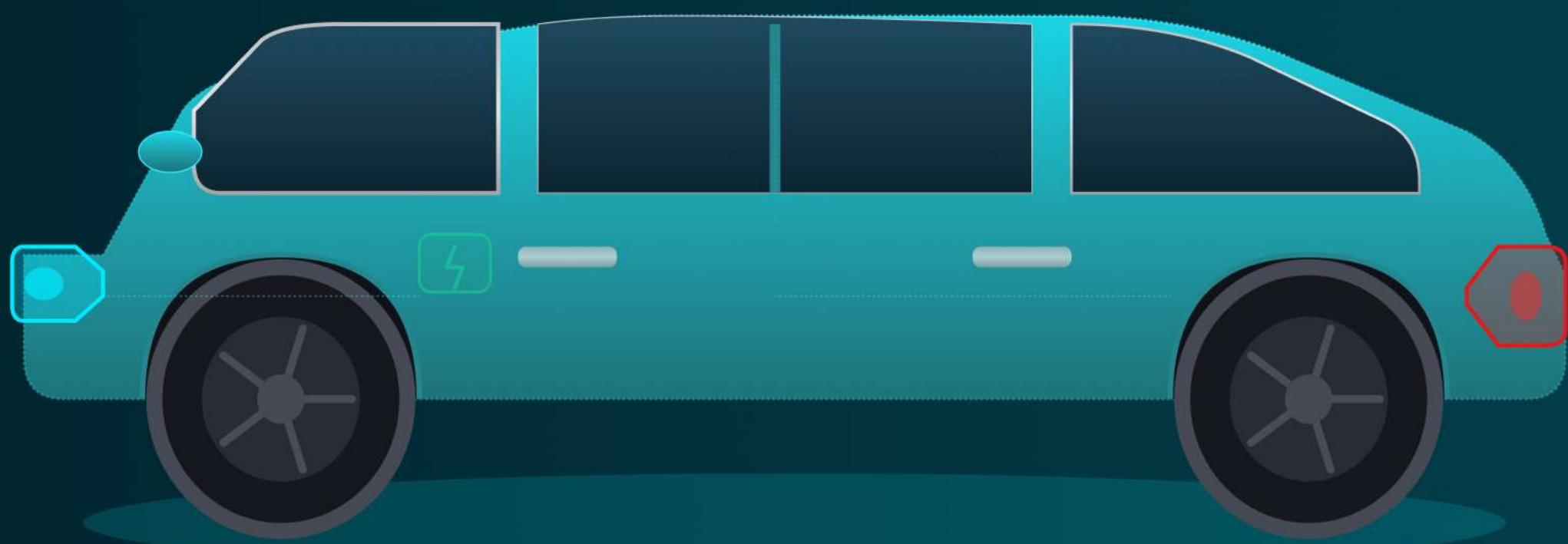
Detect issues early and guide users before problems arise.



Confidence

Build trust through accurate insights and reliable information.

Final UI Screens



01 HOME

EVCare

HomeDashboardInsightsChargingTipsInfotainment

Your EV's Personal Health Monitor

Understand Your EV.
Take Better Care.

Monitor battery health, optimize charging habits, and extend your electric vehicle's lifespan with intelligent insights.

View Vehicle Health →

Learn EV Care

02 DASHBOARD

EVCare

HomeDashboardInsightsChargingTipsInfotainment

Battery Status

78%

Current Charge78%Range Available224 mi

Health Metrics

94%

287 mi

72°

Overall HealthRange (mi)Temp (°F)

Efficiency Score

4.2

mi/kWh

Your driving efficiency is 12% above average for your vehicle model.

Next Service

45 days away

Tire rotation and cabin air filter replacement recommended.

03 CHARGING

EVCare

HomeDashboardInsightsChargingTipsInfotainment

Charging Behavior

Understand and optimize your charging patterns

Total Energy

847 kWh

This month

Charging Sessions

28

This month

Avg. Session

2.4 hrs

Per charge

Fast Charge Ratio

15%

Of total charges

Charging History

Weekly Summary

Fast Charges4Slow Charges23

Total kWh190Est. Cost\$28

04 INSIGHTS

EVCare

HomeDashboardInsightsChargingTipsInfotainment

Smart Insights

AI-powered recommendations to optimize your EV experience

This Month's Overview

Insights Generated24Actions Taken18Range Saved+42 miMoney Saved\$67

Frequent Fast Charging Detected

You've used fast charging 8 times this month.

Excellent Driving Efficiency

Your average energy consumption is 4.2 mi/kWh, which is 12% better than similar vehicles.

Optimal Charging Window

Charging between 11 PM - 6 AM could save you up to \$15/month in energy costs.

Battery Preconditioning Active

Automatically warms the battery in cold weather for better range and charging speed.

Regenerative Braking Maximized

You recovered 87 kWh this month through regenerative braking.

Explore All Insights →

05 TIPS

EVCare

HomeDashboardInsightsChargingTipsInfotainment

EV Care Tips

Simple habits to extend your battery life and maximize range

Keep Battery Between 20-80%

Avoid Extreme Temperatures

Prefer Slow Charging

Drive Smoothly

Mind Your Speed

Precondition in Cold Weather

Use Cabin Preconditioning

Check Tire Pressure Monthly

Schedule Charging During Off-Peak

Update Software Regularly

Get Personalized Tips →

06 INFOTAINMENT

10:09 PM

N N

24°

Navigation

Traffic is low, chance of rain 73%. 32 minutes to destination.

Arrived at 12:45 PM

Weather

Partly cloudy

Low 71° - High 84° W at 15 km/h

Now Playing

Ambient Drive

Chris Brown

Battery

87

Range

240

Speed

45

Charging

Fast

150 kW available

Efficiency

94%

Eco mode active

Call

POI

Charge

A/C

EVCare

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01 HOME

02 DASHBOARD

03 CHARGING

04 INSIGHTS

05 TIPS

06 INFOTAINMENT

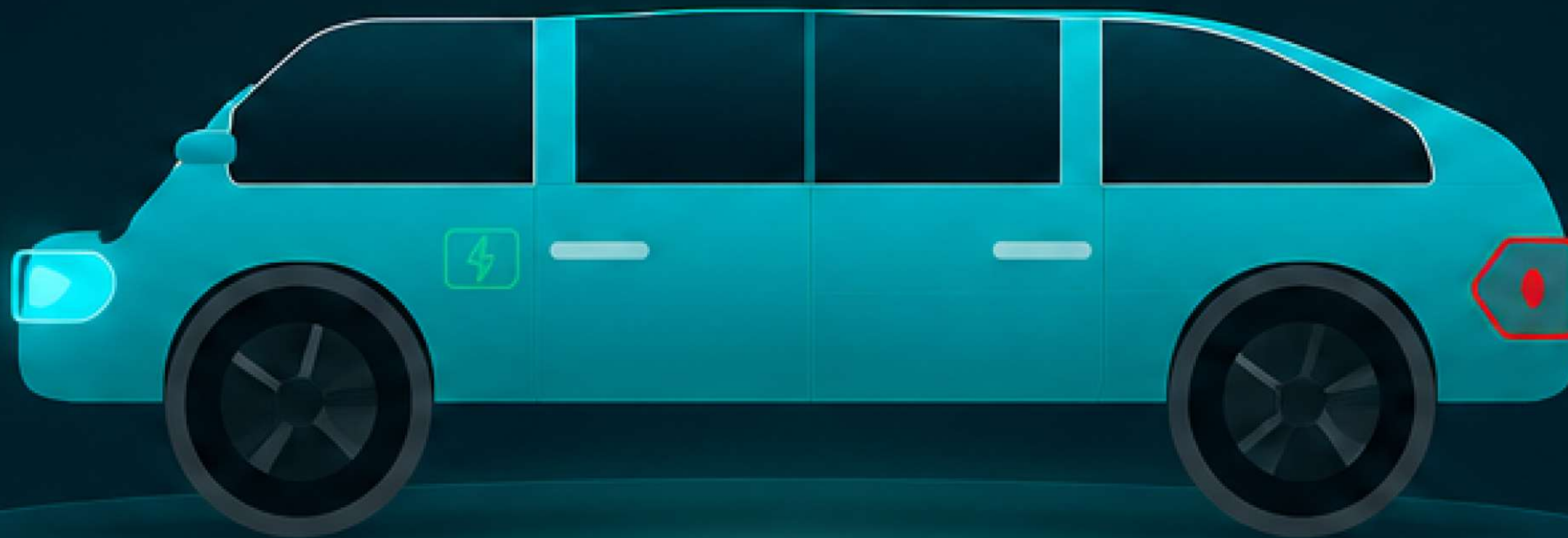
THANK YOU



Thank you for exploring the **EVCare** case study.
I hope you enjoyed the journey!



Smarter Care. Longer Drive.



Designed by
Priyadharshini R
UI/UX Designer



Empowering EV owners with the insights they need to drive **smarter**, **charge better**, and go further.